Biology Chapters B1–B5 — Cambridge IGCSE Biology Study Guide — Cheat Sheet

Key Terms

Organism	A living thing, such as a human, a platypus, a bacterium or a mango tree
Cell	The smallest unit from which all organisms are made
Unicellular	Made of just a single cell; bacteria and yeast are unicellular
Multicellular	Made of many cells; humans and plants are multicellular
Movement	An action by an organism or part of an organism causing a change of position or place
Respiration	The chemical reactions in cells that break down glucose and other nutrient molecules to release energy for the cell to use
Sensitivity	The ability to detect changes in the internal or external environment and respond to them
Growth	A permanent increase in size and dry mass, by the growth of cells and the addition of new cells
Reproduction	The processes that make new organisms of the same species
Excretion	The removal from the body of the waste products of metabolism (unwanted or toxic substances) and of substances in excess of requirements
Nutrition	Taking in substances from the environment and using them for energy, growth and development
Metabolism	All the chemical reactions that take place in an organism's cells, including respiration
Metabolic reactions	Chemical reactions that take place in living organisms
Light microscope	A microscope that shines light through the specimen and uses glass lenses to magnify and focus the image; a very good one magnifies about ×1500
Electron microscope	A microscope that can magnify far more than a light microscope; needed to see ribosomes, and to see mitochondria clearly
Photomicrograph	A photograph taken using a light microscope
Electron micrograph	A picture taken using an electron microscope, such as Figure B1.10
Cell membrane	A very thin layer of protein and fat surrounding the cytoplasm of every cell; it controls what enters and leaves the cell
Partially permeable	Allows some molecules and ions to pass through, but not others
Cell wall	A tough layer outside the cell membrane; found in the cells of plants, fungi and bacteria
Cellulose	A polysaccharide that forms criss-crossing fibres; plant cell walls are made mainly of cellulose
Fully permeable	Allows all molecules and ions to pass through it
Cytoplasm	The jelly-like material that fills a cell; it is nearly all water, with many substances (especially proteins) dissolved in it
Vacuole	A fluid-filled space inside a cell, separated from the cytoplasm by a membrane
Cell sap	The fluid that fills the large vacuoles in plant cells; a solution of sugars and other substances
Vesicle	A very small vacuole
Nucleus	A structure containing DNA in the form of chromosomes (plural: nuclei)
Chromosome	A length of DNA, found in the nucleus of a cell; it contains genetic information in the form of many different genes
DNA	A molecule that contains genetic information, in the form of genes, that controls the proteins that are made in the cell
Chloroplasts	Small structures found inside some plant cells, inside which photosynthesis takes place
Chlorophyll	The green pigment inside chloroplasts that absorbs energy from sunlight for photosynthesis
Starch grains	Structures that consist of thousands of starch molecules, stored in some plant cells
Ribosome	A very small structure in a cell that uses information on DNA to make protein molecules
Mitochondrion	A small structure in a cell where aerobic respiration releases energy from glucose (plural: mitochondria)
Aerobic respiration	Chemical reactions in mitochondria that use oxygen to break down glucose and other nutrient molecules to release energy for the cell to use
Bacteria	Unicellular organisms whose cells do not contain a nucleus (singular: bacterium)
Prokaryotic cells	Cells with no nucleus; bacteria have prokaryotic cells
Circular DNA	The single large circle of DNA found in a bacterial cell instead of chromosomes in a nucleus; sometimes called a bacterial chromosome
Plasmids	Small, circular molecules of DNA found in many prokaryotic cells in addition to the main, much larger circle of DNA
Peptidoglycan	The substance that bacterial cell walls are made of, instead of cellulose
Specialised cell	A cell whose structure is modified to help it carry out a particular function effectively
Tissue	A group of similar cells that work together to perform a particular function
Epidermis	The outer layer of tissue on a plant; also the outer layer of an animal's skin
Organ	A structure made up of a group of tissues, working together to perform specific functions
Organ system	A group of organs with related functions, working together to perform body functions
Digestion	The breakdown of food
Magnification	How many times larger an image is than the actual object
Micrometre (μm)	A unit of length equal to one thousandth of a millimetre; 1 mm = 1000 μm and 1 μm = 1×10^{-6} m
Particles	(In this context) the smallest pieces of which a substance is made; particles can be molecules, atoms or ions
Diffusion	The net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement
Net movement	Overall or average movement
Concentration gradient	An imaginary 'slope' from a high concentration to a low concentration
Random movement	Movement in any direction, purely by chance; particles travel in straight lines and change direction when they collide with other particles
Kinetic energy	The energy of movement; the energy for diffusion and osmosis comes from the kinetic energy of the moving particles, not from respiration
Rate of diffusion	How fast particles diffuse; it is faster at a higher temperature, with a larger surface area, a steeper concentration gradient and a shorter diffusion distance
Diffusion distance	The distance particles have to travel, e.g. the thickness of a membrane; the shorter the distance, the faster the diffusion
Surface area to volume ratio	The surface area of an object divided by its volume; cutting a cube into smaller cubes increases this ratio, so substances diffuse in faster
Independent variable	The variable that you change in an experiment
Dependent variable	The variable that you measure as you collect your results
Standardised variable	A variable that you keep the same throughout an experiment so that it cannot affect the results (also called a control variable)
Solvent	The liquid in which a substance dissolves; in the solutions in and around cells, the solvent is water
Solute	A substance that is dissolved in a solvent, e.g. sugar dissolved in water
Partially permeable membrane	A membrane that lets some molecules through (e.g. small water molecules) but not others (e.g. much larger sugar molecules); cell membranes are partially permeable
Osmosis	The diffusion of water molecules through a partially permeable membrane

Osmosis (in terms of water potential)	The net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane
Water potential	A way of describing how much water a solution contains compared with dissolved solute; a dilute solution has a high water potential, a concentrated solution has a low water potential, and pure water has the highest water potential of all
High water potential	An area where there are a lot of water molecules, that is, a dilute solution
Low water potential	An area where there are not many water molecules, that is, a concentrated solution
Water potential gradient	A difference in water potential between two areas; water moves down it, from a high to a low water potential
Turgid	A description of a plant cell that is tight and firm
Turgor pressure	The pressure of the water pushing outwards on a plant cell wall
Flaccid	A description of a plant cell that has lost its turgor pressure and is soft
Plasmolysed	A description of a cell in which the cell membrane tears away from the cell wall
Plasmolysis	What happens when a plant cell loses so much water by osmosis that its cell membrane tears away from the cell wall; it can kill the cell
Wilting	The drooping of a plant that happens when its cells become flaccid and the plant loses its firmness
Active transport	The movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration
Carrier proteins (protein carriers)	Protein molecules in cell membranes that can use energy to change shape and move ions or molecules into or out of a cell
Root hair cell	A cell on the surface of a plant root that takes in water and mineral ions from the soil; it uses active transport to take in nitrate ions
Dialysis	A treatment for kidney failure in which unwanted substances diffuse out of the blood, through a very thin membrane, into dialysis fluid that then flows away
Indicator	A dye that changes colour depending on pH; cresol red is red-purple in alkaline conditions and turns yellow when acid is added
Anomalous result	A result that does not fit the pattern or trend shown by the other results
Biological molecules	Molecules that make up living cells, such as carbohydrates, fats and proteins; they are not often found in non-living things
Element	A substance made of only one kind of atom, e.g. carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
Compound	A substance made of two or more different elements chemically combined, e.g. glucose
Molecule	A particle made up of two or more atoms combined together, e.g. O_2 or $C_6H_{12}O_6$
Carbohydrates	Substances that include sugars, starch and cellulose; they contain carbon, hydrogen and oxygen, with about twice as many hydrogen atoms as carbon or oxygen atoms
Sugars	Carbohydrates that have relatively small molecules; they are soluble in water and they taste sweet
Glucose	A sugar that is used in respiration to release energy; formula $C_6H_{12}O_6$; the form in which carbohydrate is transported in the blood
Reducing sugar	A sugar, such as glucose, which turns Benedict's solution orange-red when they are heated together
Starch	A carbohydrate that is used as an energy store in plant cells; made of many glucose molecules in a long chain that coils into a spiral
Glycogen	A carbohydrate that is used as an energy store in animal cells; made of many glucose molecules in branched chains
Lipids	Substances containing carbon, hydrogen and oxygen; they are insoluble in water and are used as energy stores in organisms; another name for fats and oils
Fats	Lipids that are solid at room temperature
Oils	Lipids that are liquid at room temperature

Glycerol	One of the two kinds of smaller molecule that make up a fat; each fat molecule contains one glycerol molecule
Fatty acids	Smaller molecules that combine with glycerol to make fats and oils; each fat molecule contains three fatty acid molecules
Protein	A substance whose molecules are made of many amino acids linked together; each different protein has a different sequence of amino acids
Amino acids	Substances with molecules containing carbon, hydrogen, oxygen and nitrogen; there are 20 different amino acids found in organisms
Enzymes	Proteins that act as biological catalysts, speeding up chemical reactions in living organisms; all enzymes are proteins
Antibodies	Molecules secreted by white blood cells, which bind to pathogens and help to destroy them; antibodies are proteins
Pathogens	Microorganisms, such as bacteria, that cause disease
Haemoglobin	The red pigment that transports oxygen in mammalian blood; it is a protein
Keratin	The protein that forms hair (and fingernails)
Iodine solution	A solution of iodine in potassium iodide; it is orange-brown, and turns blue-black when mixed with starch
Benedict's solution	A blue liquid that turns orange-red when heated with a reducing sugar
Biuret reagent	A blue solution (also called biuret solution) that turns violet (purple) when mixed with proteins
Emulsion	A liquid containing two substances that do not fully mix; one of them forms tiny droplets dispersed throughout the other
Iodine test	The food test for starch: add iodine solution to the food; a blue-black colour means starch is present
Benedict's test	The food test for reducing sugars: heat the food with Benedict's solution in a water-bath; green, yellow or orange-red means reducing sugar is present
Biuret test	The food test for proteins: add biuret solution to the food and shake, with no heating; a violet (purple) colour means protein is present
Ethanol emulsion test	The food test for fats and oils: shake the food with ethanol, then pour the ethanol into water; a milky white emulsion means fat or oil is present
Water-bath	A container of water kept at a steady temperature, used to heat test-tubes gently and evenly, e.g. at about 80 °C for the Benedict's test
Catalyst	A substance that increases the rate of a chemical reaction and is not changed by the reaction
Biological catalyst	A catalyst made by living cells; enzymes are biological catalysts that speed up the reactions inside organisms
Substrate	The substance that an enzyme causes to react (for example, starch is the substrate of amylase)
Product	The new substance formed by a chemical reaction
Active site	The part of an enzyme molecule to which the substrate temporarily binds
Complementary	With a perfect mirror-image shape; the active site and its substrate have complementary shapes, so they fit together
Enzyme–substrate complex	The short-lived structure formed as the substrate binds temporarily to the active site of an enzyme
Specificity	Of enzymes, only able to act on a particular (specific) substrate
Lock and key model	The idea that a substrate fits an enzyme's active site like a key fits a lock; only a substrate with the complementary shape can fit
Carbohydrases	Enzymes that break down carbohydrates (for example, amylase, maltase, sucrase and lactase)
Protease	An enzyme that catalyses the breakdown of protein to amino acids
Lipases	Enzymes that break down lipids (fats and oils) into fatty acids and glycerol
Amylase	An enzyme that catalyses the breakdown of starch to maltose
Maltase	An enzyme that catalyses the breakdown of maltose to glucose
Sucrase	An enzyme that breaks down sucrose

Lactase	An enzyme that breaks down the milk sugar lactose into smaller sugar molecules that can be absorbed into the blood
Catalase	An enzyme that catalyses the breakdown of hydrogen peroxide to water and oxygen; it works inside all living cells
Hydrogen peroxide	A very dangerous substance made by many chemical reactions inside cells; catalase must break it down immediately
Lactose intolerance	Feeling ill (diarrhoea, abdominal pain, flatulence) after eating dairy foods, because the body does not make lactase and lactose is not broken down
Optimum	Best; for example, the optimum temperature of an enzyme is the temperature at which its activity is greatest
Optimum temperature	The temperature at which an enzyme works fastest; about 37 °C for enzymes from the human body
Optimum pH	The pH at which an enzyme works fastest; around pH 7 for most enzymes, but about pH 2 for a protease in the stomach
Denaturation	Damage to an enzyme's shape (including its active site) caused by high temperature or an unsuitable pH; the enzyme is denatured and can no longer bind its substrate or catalyse its reaction
Effective collision	A collision in which a substrate molecule enters an enzyme's active site, forms an enzyme–substrate complex and is converted to product
Range	The lowest to the highest value
Buffer solution	A liquid that has a known pH, and keeps that pH steady all the time, even if chemical reactions take place
Thymolphthalein	An indicator that is blue in alkaline conditions and loses its colour when the pH becomes less alkaline
Photosynthesis	The process by which plants synthesise carbohydrates from raw materials using energy from light
Organic substances	Carbon-containing biological molecules such as carbohydrates, fats and proteins; plants can make them but animals must take them in as food
Raw materials of photosynthesis	Carbon dioxide (from the air) and water (from the soil), which react to make glucose and oxygen
Chloroplast	The part of a plant cell where photosynthesis happens; contains chlorophyll (spread out on membranes) and the enzymes for photosynthesis
Sucrose	A sugar whose molecules are made of glucose and another similar molecule (fructose) linked together; the form in which plants transport carbohydrate
Nitrate ions	Mineral ions absorbed from the soil through the root hairs; they supply the nitrogen needed to make amino acids, proteins and chlorophyll
Magnesium ions	Mineral ions absorbed from the soil; magnesium is needed to make chlorophyll
Nectar	A sweet liquid secreted by many insect-pollinated flowers, to attract their pollinators
Pollen grains	Small structures which contain the male gametes of a flower
Pollinator	An animal (insect, bat or bird) that carries pollen grains from one flower to another
Xylem vessels	Tubes that carry water and mineral ions from the roots up to the leaves
Phloem tubes	Tubes that carry sucrose (and amino acids) from one part of a plant to another
Vascular bundles	Collections of xylem tubes and phloem vessels running side by side, which form the veins in a leaf
Lamina	The broad, flat part (blade) of a leaf
Midrib	The main vein running down the centre of a leaf
Leaf stalk	The stalk that joins the lamina to the rest of the plant; it contains vascular bundles
Dicotyledonous (dicot) plant	A flowering plant whose leaves usually have a branching network of veins
Transverse section (TS)	A thin slice cut across a structure, such as a leaf, so that its layers of cells can be seen under a microscope
Cuticle	A thin, transparent, waterproof covering made of a waxy substance, on the upper surface of a leaf
Upper epidermis	The upper layer of tightly packed, transparent cells that reduces water vapour loss and lets light through
Palisade mesophyll	A layer of tall, narrow cells with very large numbers of chloroplasts near the top of the leaf; the main site of photosynthesis
Spongy mesophyll	A layer of photosynthesising cells with many air spaces between them, allowing gases to diffuse through the leaf
Air spaces	Gaps between spongy mesophyll cells through which carbon dioxide, oxygen and water vapour diffuse
Lower epidermis	The bottom layer of cells of a leaf; it contains the stomata and usually has no cuticle
Stoma (plural stomata)	An opening in the lower epidermis that lets carbon dioxide and oxygen diffuse in and out, and water vapour diffuse out
Guard cells	A pair of cells (containing chloroplasts) surrounding each stoma; they change shape to open and close it
Variegated leaf	A leaf that has some white areas (no chlorophyll) and some green areas (with chlorophyll)
Destarching	Leaving a plant in the dark for long enough for it to use up its starch stores
Control	A standard sample that you use as a comparison, to find the effect of changing a variable
Potassium hydroxide solution	A solution that absorbs carbon dioxide; used to remove carbon dioxide from the air around a leaf
Sodium hydrogencarbonate solution	A solution that provides carbon dioxide for aquatic plants; a higher concentration supplies more carbon dioxide
Rate of photosynthesis	How much photosynthesis happens in unit time, e.g. the number of oxygen bubbles released by an aquatic plant in one minute
Limiting factor	The factor in shortest supply (light intensity, carbon dioxide concentration or temperature) that stops the rate of photosynthesis going any higher
Heat shield	A transparent barrier between a lamp and a plant that lets light through but stops the lamp's heat changing the temperature
Glasshouse	A glass building in which temperature, light and carbon dioxide can be controlled so that crops photosynthesise quickly and give a high yield
Biofuels	Fuels made from plants grown for the purpose, which can be burnt to produce electricity or move vehicles

Key Equations

- cells → tissues → organs → organ systems → organism
- 11. Levels of organisation: cells → tissues → organs → organ systems → organism
- 13. Aerobic respiration (in mitochondria, details in Chapter B9): glucose + oxygen → carbon dioxide + water, releasing energy
- 1. Diffusion: higher concentration → lower concentration (down a concentration gradient) by random movement; the energy is the particles' kinetic energy, not respiration
- 5. Osmosis: higher water potential (dilute solution) → lower water potential (concentrated solution), through a partially permeable membrane
- 9. Active transport: lower concentration → higher concentration (against a concentration gradient), through carrier proteins in the cell membrane, using energy from respiration
- - many glucose molecules → starch (the energy store in plants)
- - many glucose molecules → glycogen (the energy store in animals)
- Iodine test: orange-brown → blue-black means starch is present.
- Benedict's test: blue → green → yellow → orange-red as the amount of reducing sugar increases.
- 1 glycerol + 3 fatty acids → 1 fat (lipid) molecule
- Ethanol emulsion test: clear → milky white emulsion means fat or oil is present.
- Many amino acids (20 kinds) joined in a specific sequence → one protein molecule
- Biuret test: blue → violet (purple) means protein is present.
- 5. Many glucose molecules → starch (energy store in plants) or glycogen (energy store in animals)
- 6. 1 glycerol + 3 fatty acids → 1 fat or oil (lipid) molecule
- 7. Many amino acids (20 kinds) in a specific sequence → a protein; a different sequence gives a different protein
- 8. Iodine test for starch (no heating): orange-brown → blue-black
- 9. Benedict's test for reducing sugars (heat in a water-bath at about 80 °C): blue → green → yellow → orange-red as the concentration of reducing sugar increases
- 10. Biuret test for proteins (no heating): blue → violet (purple)

Key Facts

- Q: The list below contains some features of living organisms: breathing, excretion, a blood system, a nervous system, sensitivity, growth,

reproduction, movement, nutrition, respiration. Which of these features are found in all living organisms?

- A: Excretion, sensitivity, growth, reproduction, movement, nutrition and respiration. These are the seven characteristics of living things. Breathing, a blood system and a nervous system are not found in all organisms: plants and bacteria do not breathe (breathing is not the same as respiration), and they have no blood system and no nervous system.
- Q: If you have a microscope, how can you distinguish between a cell from a plant, and a cell from an animal?
- A: Look for a cell wall. All plant cells have a cell wall outside the cell membrane, but animal cells never do. You could also look for chloroplasts and a large central vacuole: if a cell has chloroplasts, it is a plant cell. However, a cell without chloroplasts could still be a plant cell (for example, a root cell), so the cell wall is the most reliable clue.
- Q: Can you think of any advantages for larger organisms of consisting of multiple cells, rather than just one larger cell?
- A: In a multicellular organism, different cells can become specialised to carry out particular functions (for example, nerve cells, muscle cells and red blood cells), so the organism can do many different jobs efficiently. Other advantages: if some cells are damaged or die, they can be replaced and the organism survives; and small cells can exchange substances such as oxygen with their surroundings quickly, whereas in one giant cell substances would take too long to reach the middle.
- Aim: To learn the seven characteristics of living things and what each one means, by playing a memory card game.
- Group size: A group of four or five.
- You will need: 14 pieces of blank card, all exactly the same.
- Expected results: Every name card is eventually paired with its correct description. Players who remember where cards are, and who know the descriptions well, collect the most pairs.
- Conclusion: By the end you should be able to name all seven characteristics and match each one to its meaning. The cards must be identical so that nobody can recognise a card from its back.
- Safety: There are no special hazards in this activity.
- Figure B1.03 (an animal cell): a cell with an irregular, rounded outline. It shows a thin cell membrane, cytoplasm, a large dark nucleus, several mitochondria, tiny dots labelled ribosomes, and a small vacuole or vesicle.
- Figure B1.04 (a plant cell): a box-shaped cell. It shows a cell wall on the outside with the cell membrane pressed against it, a thin layer of cytoplasm around the edge containing the nucleus, green chloroplasts, mitochondria and ribosomes, and a very large vacuole filling most of the middle of the cell.
- Figure B1.05: a photomicrograph of cells from the trachea (windpipe) of a mammal. The cells have been stained (coloured) with a dye that makes the nucleus look darker than the cytoplasm. The cell membrane, cytoplasm and nucleus are labelled.
- Figure B1.06: a photomicrograph of cells from a moss plant. The cell walls form a clear network, and each cell contains many green chloroplasts and a nucleus. You cannot see the cell membranes, because they are pressed tightly against the inside of the cell walls.
- Plant cells have very large, permanent vacuoles, which contain a solution of sugars and other substances called cell sap. When the vacuole is full, it presses outwards on the rest of the cell and helps it to keep its shape.
- Animal cells have much smaller vacuoles, called vesicles, which also contain solutions.
- Key idea: plant cells have everything an animal cell has, plus a cell wall, a large permanent vacuole and (often) chloroplasts.
- Bacteria (singular: bacterium) are unicellular organisms. Bacterial cells are rather different from the cells of animals and plants. Figure B1.12 is a diagram of a bacterium: a capsule-shaped cell with labels for the cell wall (made of peptidoglycan, not cellulose), cell membrane, cytoplasm, ribosomes, circular DNA and plasmids.
- Circular DNA. Instead of chromosomes inside a nucleus, bacteria have a circle of DNA, sometimes called a bacterial chromosome. The DNA has exactly the same function as in other cells: it provides instructions for making proteins.
- Cell wall. Bacterial cells always have a cell wall. Unlike plant cells, this cell wall is not made of cellulose. But its function is the same as in plant cells: it helps to support the cell, and prevents it from bursting if the cell takes up a lot of water.
- Cell wall: a thick outer layer, often drawn as a double line, around plant and bacterial cells. An animal cell has only a thin single line (its membrane) around it.
- Cell membrane: a thin line around the cytoplasm. In plant cells it lies right against the inside of the wall, so it is hard to see.
- Nucleus: a large round or oval body, usually the darkest part of a stained cell.

Quick Q&A

Question	Answer
How many micrometres are there in 1 mm?	There are 1000 µm in 1 mm. To convert millimetres to micrometres, multiply by 1000.
The diagram below shows a cell from a plant leaf. Identify which labelled part contains DNA. (1 mark) A) the outer double line that forms the boundar	B — the small oval structure is the nucleus, which contains DNA in the form of chromosomes. (A is the cell wall, C is the cytoplasm and D is the large
A diagram of a flower is 8 cm across. The magnification of the diagram is ×4. Identify the actual width of the flower. (1 mark) A) 3.2 mm B) 20 mm C)	B — 20 mm. Actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm, and 2 cm × 10 = 20 mm. D (32 cm) comes from multiplying instead of dividing,
These questions are about the diagram in Figure B2.04, which shows a dilute sugar solution (left) separated from a concentrated sugar solution (right)	The solvent is water. The solute is sugar.
Copy and complete this sentence: When an animal cell is placed in pure water, water …… the cell by …… through the partially permeable cell ……	When an animal cell is placed in pure water, water enters (moves into) the cell by osmosis through the partially permeable cell membrane.
You read about the enzyme lactase at the start of this chapter. a) What type of enzyme is lactase? b) Name the substrate of lactase.	a) Lactase is a carbohydrase, because it breaks down a carbohydrate (the sugar lactose) into smaller sugar molecules. b) Lactose, the sugar found in
The figure below shows some stages in an enzyme-controlled reaction. (4 marks) (The figure shows three stages, from top to bottom. Stage 1: an enzyme	P = substrate [1] Q = active site [1] R = enzyme–substrate complex [1] S = products (the product molecules) [1]
Animals do not make or store starch. What is the substance that animals store, which is made of chains of glucose molecules?	Glycogen (stored mainly in the liver and muscles).